# Income Inequality

## Contents



# 1 Introduction

In this part of the course you will learn how to define and measure income inequality and poverty. The focus is on analyzing the evolution of inequality and poverty by individual characteristics in Canada and to compare Canada with other countries. Also, you will analyze the comovement between inequality, poverty and real per capita GDP.

### Key Terms

- **Decile:** When the population is divided into ten equal groups ordered with respect to a variable, the groups are called deciles. For example, the first income decile is one tenth (10%) of the population with the lowest income.
- **Economic growth:** This is the growth rate of the real per capital GDP.
- **Economic mobility:** This characterizes the ability for individuals to move from one part of the income distribution to another.
- **Gini coefficient:** This is a global measure of inequality. It goes from 0 (perfect equality) to 1 (perfect inequality).
- **Income share:** This is the proportion of total income that goes to an individual or a group. For example, if the income share of females is 55%, it means that 55% of the total income is earned by women.
- **Lorenz Curve:** This is a line chart that illustrates the cumulative income share of the population as a function of the proportion of the population. For example, the points (0.05,0.10) and (0.60, 0.50) on the Lorenz curve mean that 10% of the poorest individuals share 5% of total income of the economy and 50% share 60% of total income.

- **Low income measure (LIM):** This is equal to half the median of income. It is one of the measures of the poverty line. Individuals with income less than the LIM are considered below the poverty line.

- **Market basket measure (MBM):** This is the value of a basket of goods used in one definition of the poverty line. Individuals with income less than the MBM are considered below the poverty line.

- **Median:** This is a number that separates the distribution of a variable in two equal groups. For example, the median of income in Canada is $40,000 if income is less then $40,000 for 50% of the population and it is higher for the other 50%.

- **Per capita:** This is a Latin word that means by head. For example, the per capita income is the average income by individual.

- **Quintile:** When the population is divided into five equal groups ordered with respect to a variable, the groups are called quintiles. For example, the first income quintile is one fifth (20%) of the population with the lowest income.

# 2 Income Inequality

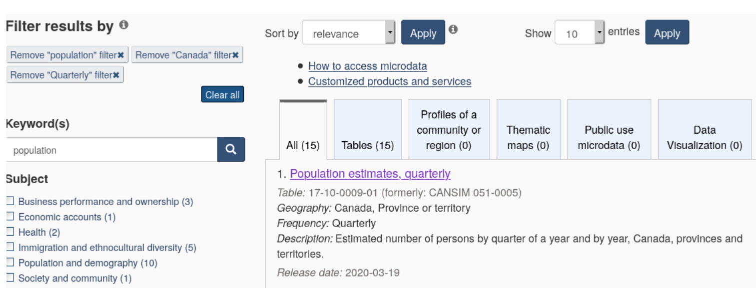
## 2.1 Economic Growth

Economic growth is defined as the growth rate of the real per capita GDP defined as

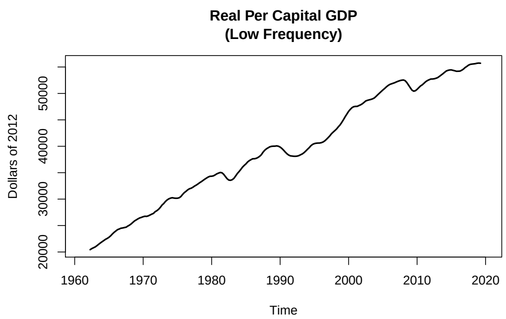
$$\boxed{\text{Real per capita GDP} = \frac{\text{Real GDP}}{\text{Population}}}\,.$$

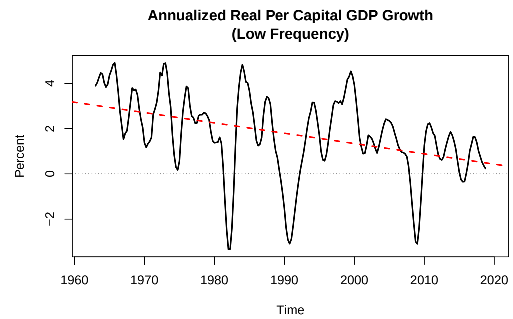
Since GDP is also a measure of total income (remember the equivalence between the three approaches to measure GDP), we also call it average real per capita income. We start by defining this measure because one of the purpose of this module is to analyze the relationship between economic growth and income inequality.

To obtain the real per capita GDP, we need to download the real GDP (Statistics Canada, 2020c) and the population (Statistics Canada, 2020e) series from Statistics Canada. The population series can be obtained by using the keyword population and selecting the Population estimates, quarterly link as in the following screenshot (Canada and Quarterly was also selected):




The following graphs show the low frequency components of the per capita real GDP (left) and the annualized per capita real GDP growth rate (right). If you try do reproduce the graphs, by careful with the units used by Statistics Canada for the GDP and population. In the tables, population is in units and GDP is in millions. Therefore you need to express population in millions by dividing the series by one million before computing the real per capita GDP. You should get, as we see on the left graph, that the real per capita GDP increased from 20,000 to 55,000 dollars of 2012 between 1961 and 2020. Since its trending behaviour is roughly constant, it implies a decreasing growth rate on average as shown by the red dashed line on the right graph.




This result implies that Canadians get richer and richer in real terms on average. Being richer in real terms means that we can buy more goods. It is therefore a sign that the standard of living of Canadians has improved over the period on average. However, the result does not imply that the standard of living of every Canadian has improved. In this module, we are interested in the allocation of real GDP across individuals, how the allocation has evolved through time and how it is related to economic growth.

## 2.2 Measuring Inequality: Income by Quintile

### Quintiles

When we divide the population into five equal groups based on the distribution of a given variable, the groups are called quintiles. The lowest quintile represents 20% of the population with the smallest values and the fifth quintile includes 20% of the population with the highest values. For example, suppose the population is composed of 15 individuals. The following table shows the age and expenditure of each individual.

**Expenditure and Age by Individual**

| Individual | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Age | 19 | 23 | 45 | 18 | 64 | 55 | 34 | 23 | 22 | 44 | 70 | 61 | 49 | 21 | 43 |
| Expenditure | 100 | 210 | 80 | 30 | 312 | 190 | 201 | 236 | 341 | 34 | 40 | 101 | 170 | 190 | 343 |

To construct quintiles according to the age variable, we first sort individuals by age and divide the population in five equal groups starting with the youngest and ending with the oldest:

**Age Quintiles**

| | First quintile | | | Second quintile | | | Third quintile | | | Fourth quintile | | | Fifth quintile | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Individual | 4 | 1 | 14 | 9 | 2 | 8 | 7 | 15 | 10 | 3 | 13 | 6 | 12 | 5 | 11 |
| Age | 18 | 19 | 21 | 22 | 23 | 23 | 34 | 43 | 44 | 45 | 49 | 55 | 61 | 64 | 70 |
| Expenditure | 30 | 100 | 190 | 341 | 210 | 236 | 201 | 343 | 34 | 80 | 170 | 190 | 101 | 312 | 40 |

The individuals that we put in each quintile depends on the variable. For example, we can see in the following table that if the quintiles are based on expenditure, they include different individuals.

**Expenditure Quintiles**

| | First quintile | | | Second quintile | | | Third quintile | | | Fourth quintile | | | Fifth quintile | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Individual | 4 | 10 | 11 | 3 | 1 | 12 | 13 | 14 | 6 | 7 | 2 | 8 | 5 | 9 | 15 |
| Age | 18 | 44 | 70 | 45 | 19 | 61 | 49 | 21 | 55 | 34 | 23 | 23 | 64 | 22 | 43 |
| Expenditure | 30 | 34 | 40 | 80 | 100 | 101 | 170 | 190 | 190 | 201 | 210 | 236 | 312 | 341 | 343 |

The purpose of dividing the population into quintiles is to compare the characteristics of groups of comparable individuals. For example, we may be interested in comparing the average expenditure by age quintile. The following table shows the average age and average expenditure by age quintile. The values are obtained averaging the values in each quintile. For example the average age and expenditure for the low age quintile are:

$$\text{Average age (first quintile)} = \frac{1}{3}(18 + 19 + 21) = 19.33$$

and

$$\text{Average expenditure (first quintile)} = \frac{1}{3}(30 + 100 + 190) = 106.67$$

**Average by Age Quintile**

| | First quintile | Second quintile | Third quintile | Fourth quintile | Fifth quintile |
|---|---|---|---|---|---|
| Age | 19.33 | 22.67 | 40.33 | 49.67 | 65 |
| Expenditure | 106.67 | 262.33 | 192.67 | 146.67 | 151 |

First we can compare the average age per quintile: the average age of the 20% youngest is 19.33 years and the average for the 20% oldest is 65. We can also compare other characteristics. In the table, we see that the expenditure is the highest on average in the second age quintile and the lowest in the first age quintile. Similarly, we can analyze the average by expenditure quintile:

$$\text{Average age (first quintile)} = \frac{1}{3}(18 + 44 + 70) = 44$$

and

$$\text{Average expenditure (first quintile)} = \frac{1}{3}(30 + 34 + 40) = 34.67$$

**Average by Expenditure Quintile**

| | First quintile | Second quintile | Third quintile | Fourth quintile | Fifth quintile |
|---|---|---|---|---|---|
| Age | 44 | 41.67 | 41.67 | 26.67 | 43 |
| Expenditure | 34.67 | 93.67 | 183.33 | 215.67 | 332 |

We can conclude that the average age for the 20% of the population that spend the least is 44 years and it is 43 years for the 20% of the population that spend the most. Similarly, the average expenditure for the 20% of the population that spend the least is $34.67 and it is equal to $332 for the 20% that spend the most.

Notice that we can divide the population by more than five equal groups. Other common ways of dividing the population is by deciles and percentiles. When the population is divided into ten equal groups, the groups are called deciles and when the population is divided into one hundred groups, the groups are called percentiles. For example, the bottom income decile represent 10% of the population with the lowest incomes and the top percentile is 1% of the population with the highest incomes.

### Average Income per Quintile in Canada

Dividing the population in quintiles based on income is one way to analyze the income distribution. Statistics Canada provides a table in which the number of households are divided into income quintiles, and returns several statistics by quintile (Statistics Canada, 2020b). The following table shows the average household income by income quintile (first column) and the share of total income (second column). The first column is obtained by selecting Value per household in the **Statistics** tab, the Lowest to Highest income quintiles in the **Characteristics** tab and Household disposable income in the **Income, consumption and savings** tab of the Statistics Canada table number 36-10-0587-01. The second column is obtained by replacing Value per household by Distribution of value in the **Statistics** tab (for a specific table enter the table number as keyword).

**Average Income by Quintile in 2018**

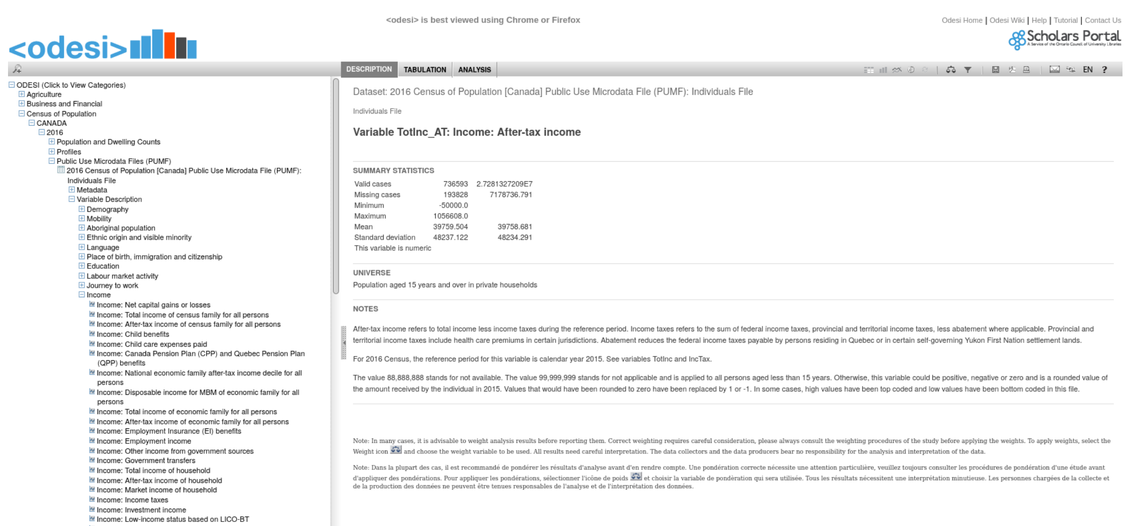
| | Average Household Income (dollars) | Share of Total Household Income (percent) |
|---|---|---|
| First Income Quintile | 23,537 | 5.90 |
| Second Income Quintile | 49,137 | 12.40 |
| Third Income Quintile | 69,943 | 17.60 |
| Fourth Income Quintile | 91,798 | 23.10 |
| Fifth Income Quintile | 162,231 | 40.90 |

Therefore, the average income for the 20% poorest households is $23,537 and it is equal to $162,231 for the 20% richest households. From the second column, it implies that 5.9% of total income is allocated to the poorest and 40.9% of total income is allocated to the richest. We can also say that 40.9% of total income goes to 20% of the population. We clearly see from this table that total total income is not allocated equally across household families.

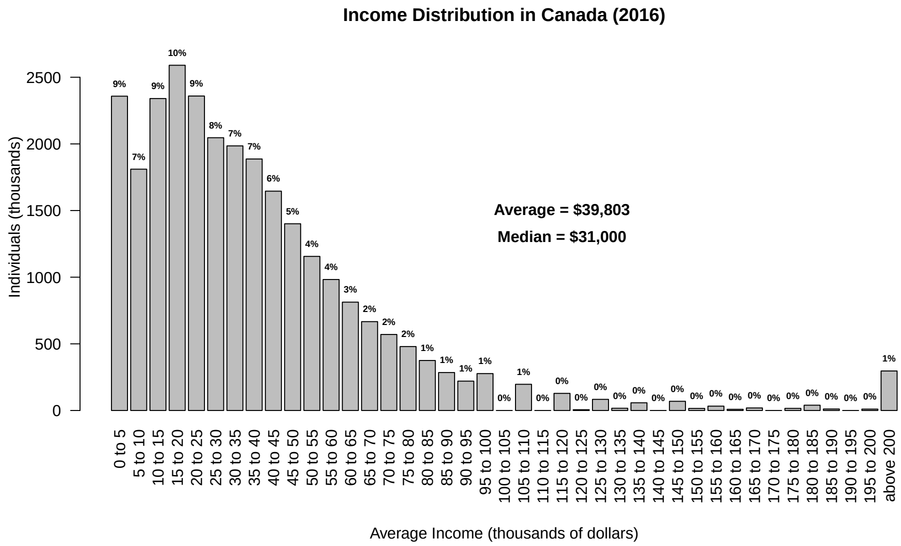
## 2.3 Measuring Inequality: Histograms

### Distribution of Income Across Individuals

To get a better picture of how total income is allocated in the population, we can create the histogram of income. Creating an histogram for income requires observations at the individuals, which is not directly available from the Statistics Canada website. We need data from the Census of population that can be downloaded from the Ontario Data Documentation, Extraction Service and Infrastructure (<odesi>). The website https://search2.odesi.ca/ is accessible publicly, but the downloads are restricted. As a student, you are allowed to download the data through the University library. If you select **ODESI Data Retrieval** in the Key Links panel on the left, you will be asked to login with your last name and Watcard barcode number. Once logged in, you can download any data files. For this course, you are not required to download the data from that source because the manipulation of survey and census data is beyond the score of the course. However, you are welcome for your own interest to surf on the website. The data used in the following is from the 2016 census (Statistics Canada, 2019). It is described on the following screenshot:



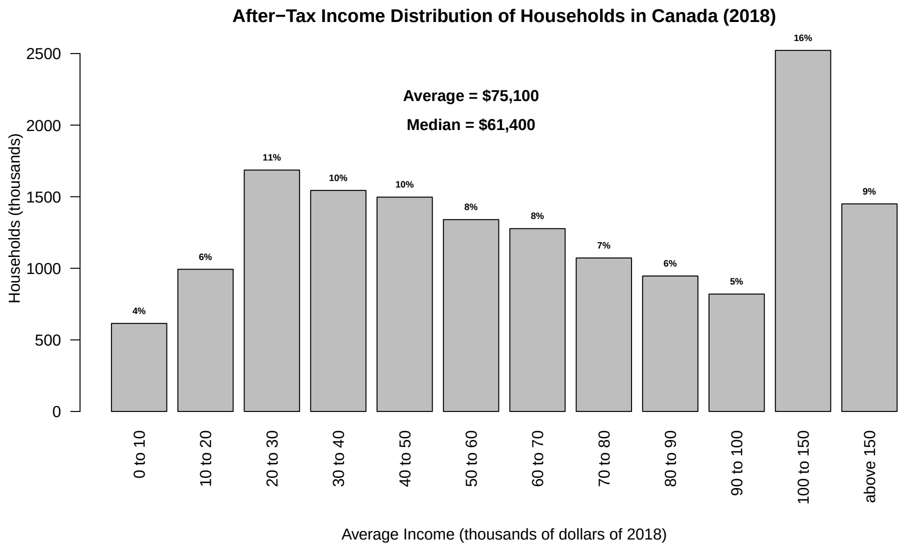

Without going into too much details, the dataset contains the after-tax income series of about 735,000 individuals. It is a representative sample that represents approximately 1/37 of the Canadian population 15 years and older. Therefore, we have to multiply the number of counts by 37 to have a sense of how many individuals we have in each income group. The following is the histogram of income from the 2016 census. The numbers on top of each bar is the percentage of individuals that belong to the income group. It is rounded to the nearest integer, so 0% means less than 0.5%.




We can see from the histogram that there is a concentration of individuals in the bottom of the distribution (low income groups): The distribution is skewed to the right with a peak for the income group $200,000 and above. The histogram shows clearly that income inequality is present in Canada.

### Distribution of Income Across Households

The above graph is the distribution of income at the individual level for the working age population. Some of the low income individuals are students working few hours per year and others are members of high income families who are voluntarily out of the labour force. For that reason, it may be more appropriate instead to look at the household income distribution because an household income includes income of all members of the household. That information is available on the Statistics Canada website, in table 11-10-0237-01 (Statistics Canada, 2020a). The following graph shows the distribution of household after-tax income in 2018. You obtain the information by selecting After-tax income in the **Income concept** tab, Economic families and persons not in an economic families in the **Economic family type** tab and all income groups in the **Statistics** tab of the table. The number on the top of each bar is the proportion of households that belong to that particular income group.




We can see that the number of income groups is smaller: we have only 12 income groups while we had 41 groups on the individual based income distribution. This is the best we can do using the formatted tables from Statistics Canada. We do, however, observe the same shape: a concentrated number of households at the bottom of the distribution (41% of households have income less than $50,000) and a peak at the top (27% of households have income greater than $100,000). Therefore, the income distribution is also unequal at the household level. Notice that this histogram also provides an imperfect picture of the income distribution because households are not of the same sizes.

## 2.4 Measuring Inequality: Average Versus Median Income

On the two graphs we also see that average income at the individual and household levels are respectively $39,803 and $75,100, and the medians are respectively $31,000 and $61,400.

### Median

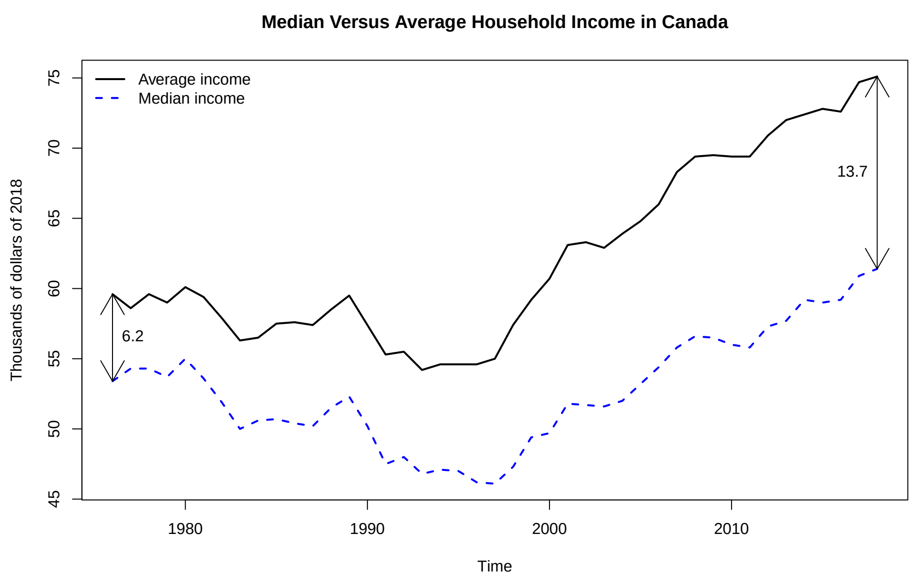
The median is the value that separates the population in two equal groups. Therefore, individual incomes are higher than $31,000 for 50% of the population and household incomes are higher than $61,400 for 50% of households. The averages are higher than the medians because many individuals and households have very high income compared with the average Canadians. The average is much more influenced by the right tail of the distribution (high income groups) than the median. To understand why, suppose Canada is composed of 4 individuals. The following table shows the income of all individuals for 2015 and 2020 with the average and median income for each year.

| Year | Individual 1 | Individual 2 | Individual 3 | Individual 4 | Average | Median |
|---|---|---|---|---|---|---|
| 2015 Income | $100 | $200 | $202 | $302 | $201 | $201 |
| 2020 Income | $100 | $200 | $202 | $400 | $225.5 | $201 |

In 2015, the median is $201 because the income is greater than $201 for 50% of the population (individuals 3 and 4) and it is less than $201 for 50% of the population (individuals 1 and 2), and the average is also $201: ($100+$200+$202+$302)/4=$201. From 2015 to 2020, the income of the fourth individual has increased to $400 while all other incomes have remained constant. Therefore, the median in 2020 is the same as in 2015 because we still have 50% of incomes on either side of \$201. However, the average has increased to \$225.5 and is higher than the median: (\$100+\$200+\$202+\$400)/4=\$225.5. Therefore, when we only increase the income of the richest individuals, the average increases and median remains unaffected. It is therefore a sign that the allocation is becoming more unequal when the gap between the average and the median is increasing.

### The Evolution of Average and Median Income in Canada

The average and median incomes in dollars of 2018 from 1976 to 2018 are available from the same Statistics Canada table by selecting Average Income and Median Income from the **Statistics** tab. The following graph plots the two series (the solid black line is the average income and the dashed blue line is the median income series). First, we see that both statistics decreased slightly on average between 1976 and the mid '90s and increased from the mid '90s to 2018. Second, the gap between the average and the median increased over the period, going from 6.2 to 13.7 thousand dollars of 2018 (see the gaps represented by the arrows). This is a sign that income inequality has increased over this period.




## 2.5 Measuring Inequality: The Gini Coefficient

Comparing inequality across regions or countries using histograms is difficult because it is not evident what we should be comparing. The purpose of the Gini coefficient is to provide one number that characterizes the level of inequality. The coefficient can be used to compare regions and periods.

### The Lorenz Curve

The Lorenz Curve is a visualization tool to analyze income inequality. To plot the curve, we need the share of total income by quintile. The following table is the same one use in previous section.

**Average Income by Quintile in 2018**

| | Average Household Income (dollars) | Share of Total Household Income (percent) |
|---|---|---|
| First Income Quintile | 23,537 | 5.90 |
| Second Income Quintile | 49,137 | 12.40 |
| Third Income Quintile | 69,943 | 17.60 |
| Fourth Income Quintile | 91,798 | 23.10 |
| Fifth Income Quintile | 162,231 | 40.90 |

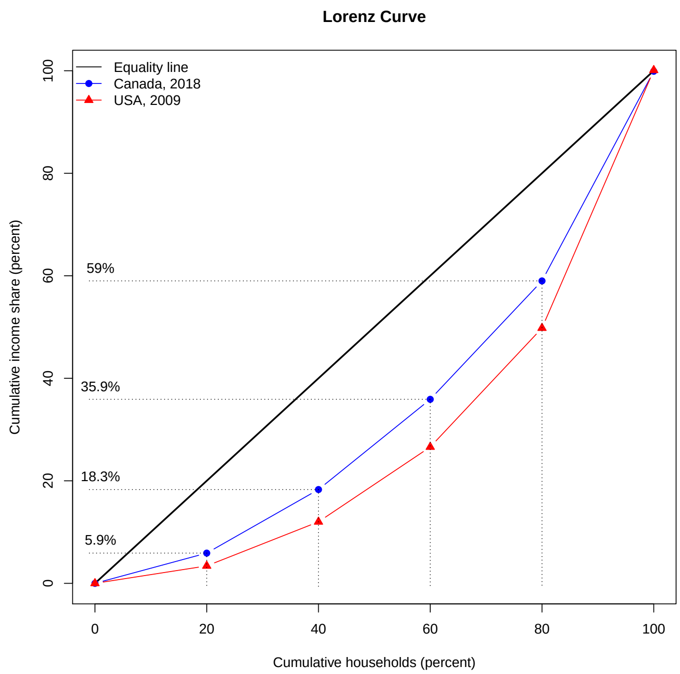
We first need the cumulative share of household income. The cumulative share of income is the share of income of the first quintile (the 20% poorest), the first two quintiles (the 40% poorest), the first three (the 60% poorest) and so on. It is the cumulative sum of the second column and is represented in the following table:

**Cumulative Income Share in Canada in 2018**

| | Cumulative Income Share (percent) |
|---|---|
| The bottom 20% | 5.90 |
| The bottom 40% | 18.30 |
| The bottom 60% | 35.90 |
| The bottom 80% | 59.00 |
| The bottom 100% | 99.90 |

The first value is 5.9, the second is 5.9+12.4=18.3, the third is 5.9+12.4+17.6=35.9 and so on. The last value is not exactly 100% because of rounding errors. The Lorenz curve, plots the values of the second column as a function of the values in the first column. The chart also includes the perfect equality curve. Income is perfectly equal if the share of total income of the bottom 20% of the distribution is 20%, it is 40% for the bottom 40% and so on.

The following graph shows the Lorenz curve for Canada in 2018 (the blue line with round markers) and for the United States in 2009 (the red line with triangular markers) using the data from the module readings. The solid black line is the perfect equality line and the coordinates are specified for Canada only for clarity. The further away the curve is from the perfect equality line, the more unequal is the income allocation. Therefore, we can conclude that the 2009 allocation in the USA is more unequal than the 2018 allocation in Canada.



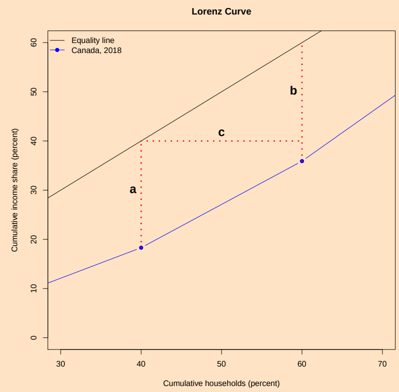

### The Gini Coefficient

The Gini coefficient is a measure based on the Lorenz curve. It is the area between the perfect equality line and the Lorenz curve divided by the area below the equality line. Since the equality line and the axes form a right triangle with a base of 1 (100%) and a height of 1 (100%), the area under the equality line is 0.5. Therefore the Gini coefficient is twice the area between the equality line and the Lorenz curve. Therefore, the coefficient is equal to 0 if the Lorenz curve is equal to the equality line and it increases as the allocation becomes more unequal. Let $h_i = (i/5 - s_i)$, for $i = 1, 2, 3, 4$, be the distance between the equality line and the Lorenz curve at 20%, 40%, 60% and 80% respectively, where $s_i$ is the cumulative income share of quintiles 1 to i. Then, the Gini coefficient is:

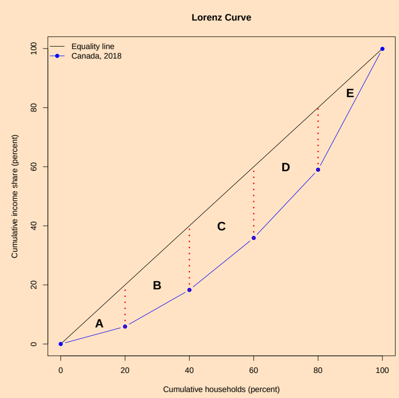
$$\text{Gini} = 0.40 \times (h_1 + h_2 + h_3 + h_4) = 0.8 - 0.4 \times (s_1 + s_2 + s_3 + s_4)$$

Using this formula, the maximum Gini coefficient is reached when all shares are equal to zero which gives a coefficient of 0.8. To have a coefficient of 1 in the most unequal allocation, we need to divide the population in more than 5 groups.

**Background Information** We can derive the Gini coefficient formula using basic geometry. If we look at the chart between two points, we get the following:




The area between the Lorenz curve and the equality curve between the two points is the area of the trapezoid with bases a and b and height c. The formula for the area of a trapezoid is (a+b)×c/2. According to our above definition a and b are respectively $h_2$ and $h_3$ and c is equal to 0.20 (20%). To compute the Gini coefficients we need to add the areas between each points and multiply the result by 2. In other words, we need to add the areas A to E in the following graph:




For A and E, we use the same formula with a=0 for A and b=0 for E. The Gini coefficient is therefore:

$$\begin{aligned}
\text{Gini} &= 2\times\left(\frac{0.20\times h_1}{2}+\frac{0.20\times(h_1+h_2)}{2}+\frac{0.20\times(h_2+h_3)}{2}+\frac{0.20\times(h_3+h_4)}{2}+\frac{0.20\times h_4}{2}\right)\\
&= 0.40\times(h_1+h_2+h_3+h_4)
\end{aligned}$$

In general, if we split the population in n equal groups, we have n-1 $h_i = (i/n - s_i)$, where $s_i$ is the cumulative income share of groups 1 to i, and c=1/n in the trapezoid area formula. Using the same logic, we can compute the Gini coefficient using the following formula:

$$\text{Gini} = \frac{2}{n}\sum_{i=1}^{n-1} h_i$$

Replacing $h_i$ by its value, we get:

$$\text{Gini} = \frac{2}{n}\sum_{i=1}^{n-1}(i/n - s_i) = \frac{n-1}{n} - \frac{2}{n}\sum_{i=1}^{n-1} s_i\,,$$

where we used the fact that $1+2+3+4+\cdots+(n-1) = n(n-1)/2$. Notice that if we divide the population in quintiles, n=5 and we get the same formula. The maximum is reached when all $s_i$ are equal to zero which implies a Gini coefficient of (n-1)/n. Therefore it approaches 1 when n gets very large, which implies a very large population in which total income is allocated to only one individual.

### Gini Coefficient in Canada and the United States

The following table shows the 2018 cumulative income share for Canada and the 2009 cumulative income share for the United States (see the module readings).

**Cumulative Income Share**

| | Canada, 2018 (percent) | USA, 2009 (percent) |
|---|---|---|
| The bottom 20% | 5.90 | 3.40 |
| The bottom 40% | 18.30 | 12.00 |
| The bottom 60% | 35.90 | 26.60 |
| The bottom 80% | 59.00 | 49.80 |
| The bottom 100% | 99.90 | 100.10 |

Once more, numbers from the last row should be equal to 100. The differences are explained by rounding error. Using the formula with shares, the Gini coefficients for Canada and the United States are respectively:

$$\text{Gini for Canada} = 0.8 - 0.4 \times (0.059 + 0.183 + 0.359 + 0.59) = 0.3236$$

and

$$\text{Gini for the USA} = 0.8 - 0.4 \times (0.034 + 0.12 + 0.266 + 0.498) = 0.4328\,.$$

According to the Gini coefficients, the allocation of income is less equal in the United States compared with Canada because the Gini coefficient is higher for the US.

### The Limit of the Gini Coefficient

As we may expect, the Gini coefficient is an imperfect measure of income inequality. We cannot expect one number to provide a complete picture of income allocation. In fact, if the sum of the income shares remains the same the coefficient is unchanged. If we consider the 2018 distribution in Canada:

**Income share in 2018**

| | Share of Total Household Income (percent) | Cumulative Income Share (percent) |
|---|---|---|
| First Income Quintile | 5.90 | 5.90 |
| Second Income Quintile | 12.40 | 18.30 |
| Third Income Quintile | 17.60 | 35.90 |
| Fourth Income Quintile | 23.10 | 59.00 |
| Fifth Income Quintile | 40.90 | 99.90 |

Suppose that income is shifted from the first to the second quintile (the poor get poorer) and from the fourth to the fifth quintile (the rich get richer) as follows:

**Alternative Income share**

| | Share of Total Household Income (percent) | Cumulative Income Share (percent) |
|---|---|---|
| First Income Quintile | 2.90 | 2.90 |
| Second Income Quintile | 15.40 | 18.30 |
| Third Income Quintile | 17.60 | 35.90 |
| Fourth Income Quintile | 15.10 | 51.00 |
| Fifth Income Quintile | 48.90 | 99.90 |

Clearly, the allocation is more unequal, but the Gini coefficient is the same:

$$\text{Gini} = 0.8 - 0.4 \times (0.034 + 0.12 + 0.266 + 0.498) = 0.4328\,.$$

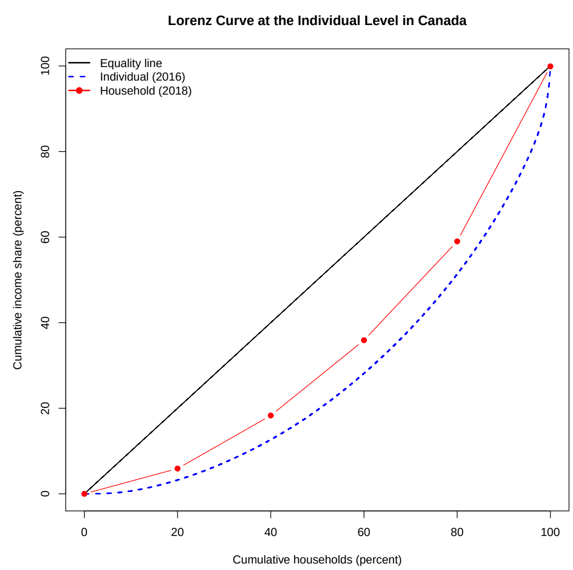
Therefore, we have to keep in mind that the Gini coefficient is one tool to compare inequalities, but it does not tell us everything about income distribution. For a complete picture, we need to dig deeper.

### Gini Coefficient in General

The Gini coefficient is not in general based on quintiles. Instead, the population is divided in many groups. It is even possible to compute the Gini coefficient using individual incomes. In that case, each individual is a group. When we divide the population into n equal groups. the formula is:

$$\text{Gini} = \frac{2}{n}\sum_{i=1}^{n-1}(i/n + s_i) = \frac{n-1}{n} - \frac{2}{n}\sum_{i=1}^{n-1} s_i\ ,$$

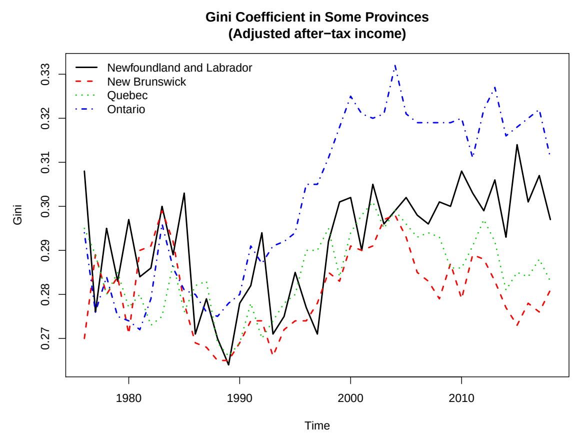
where $n$ is the number of groups (n=5 when the population is divided in quintiles) sorted by income and $s_i$ is the income share of the groups 1 to i. If we use the same data that produced the income distribution at the individual level earlier in this module, we get a Gini coefficient of 0.45. It reflects what we saw that the allocation is much more unequal at the individual level. The following graph compares the Lorenz curves at the household (red with round markers) and individual (dashed blue line) levels. We can see that the more groups we have the smoother is the Lorenz curve. It is also clear that there is more inequality at the individual level.




### The Evolution of the Gini Coefficient in Canada

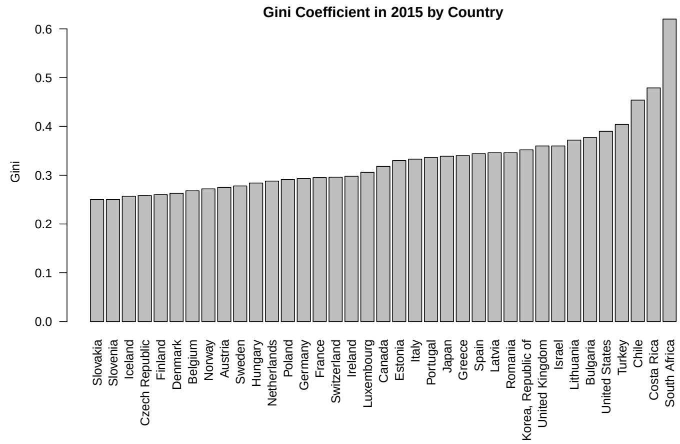
Statistics Canada publishes an annual series of the Gini coefficient that starts in 1976 (Statistics Canada, 2020d). We discussed in a previous section that the histogram of household incomes can be misleading because it does not take into account the size of the households. The Gini coefficient computed by Statistics Canada is based in household income adjusted for family size. The following is the Gini coefficient based on the adjusted after-tax income (it is income from all sources including government transfer less income tax). First, the coefficient was constant on average from 1976 to the end of the '80s, then increased from the early '90s to the early 2000s (an increase in income inequality) and start decreasing onward. We also notice that the business cycle affects inequality. For example, we can see the increase of the coefficient during the recessions of 1982 and 1990.


The following graph compares the Gini coefficient of four provinces. The Ontario province (dot-dashed blue line) is where inequality has increased the most over the period and at the end of 2018, the provinces with least income inequality was New Brunswick (dashed red) and Quebec (dotted green).




### Comparing the Gini Coefficient Across Countries

We can compare inequality across countries using the Gini coefficient, but it has to come from the same source. Coefficients from different sources may be computed differently. For example, the type of income used to calculate the coefficient may differ. The OECD published annual time series of the Gini coefficient for several countries (OECD, 2020). It is based on after-tax and after-transfer income. You can access the bar chart to compare the most recent coefficient across countries at https://data.oecd.org/chart/5VBS and the times series at https://data.oecd.org/chart/5VBV. The links offer interactive graphs that allow to navigate through different countries. You can also download the data and produce your own graphs. The following is the bar chart of the Gini coefficient in 2015 for several countries. According to the Gini coefficient, Canada is in the middle of the shown countries in terms of inequality and the United States is among the most unequal countries. Slovakia is the most equal and South Africa is the most unequal country.




## Exercises: Measuring the Gini Coefficient

1. Consider the following table

| | Average Income (dollars) |
|---|---|
| First Income Quintile | 342 |
| Second Income Quintile | 642 |
| Third Income Quintile | 844 |
| Fourth Income Quintile | 990 |
| Fifth Income Quintile | 1,053 |

   Compute the Gini coefficient.

2. Consider the following table

| | Average Income (dollars) |
|---|---|
| First Income Decile | 114 |
| Second Income Decile | 288 |
| Third Income Decile | 605 |
| Fourth Income Decile | 621 |
| Fifth Income Decile | 1,012 |
| Sixth Income Decile | 1,023 |
| Seventh Income Decile | 1,101 |
| Eighth Income Decile | 1,434 |
| Nineth Income Decile | 1,925 |
| Tenth Income Decile | 1,934 |

   Compute the Gini coefficient.

## Solution

1. First, we compute the income share. It is the average income of each quintile divided the the sum of the average incomes. For example, the income share of the first quintile is:

$$S_1 = \frac{342}{342 + 642 + 844 + 990 + 1,053} = 8.83\%$$

In the second quintile, it is equal to :

$$S_2 = \frac{642}{342 + 642 + 844 + 990 + 1,053} = 16.58\%$$

Then, we compute the cumulative income share: The first is 8.83%, the second is (8.83%+16.58%), and so on. The following table shows the shares and cumulative shares (numbers were rounded at the end)

| | Average Income (dollars) | Income share (percent) | Cumulative shares (percent) |
|---|---|---|---|
| First Income Quintile | 342 | 8.83 | 8.83 |
| Second Income Quintile | 642 | 16.58 | 25.42 |
| Third Income Quintile | 844 | 21.80 | 47.22 |
| Fourth Income Quintile | 990 | 25.57 | 72.80 |
| Fifth Income Quintile | 1,053 | 27.20 | 100.00 |

Then, we use the formula:

$$\text{Gini} = 0.8 - 0.4 \times (s_1 + s_2 + s_3 + s_4),$$

where $s_i$ are the comulative shares. The coefficient is therefore:

$$\text{Gini} = 0.8 - 0.4 \times (0.0883 + 0.2542 + 0.4722 + 0.728) = 0.18$$

2. First, we compute the income share. It is the average income of each decile divided the the sum of the average incomes. For example, the income share of the first decile is:

$$S_1 = \frac{114}{114 + 288 + 605 + 621 + 1,012 + 1,023 + 1,101 + 1,434 + 1,925 + 1,934} = 1.13\%$$

In the second decile, it is equal to :

$$S_2 = \frac{288}{114 + 288 + 605 + 621 + 1,012 + 1,023 + 1,101 + 1,434 + 1,925 + 1,934} = 2.86\%$$

Then, we compute the cumulative income share: The first is 1.13%, the second is (1.13%+2.86%), and so on. The following table shows the shares and cumulative shares (numbers were rounded at the end)

| | Average Income (dollars) | Income share (percent) | Cumulative shares (percent) |
|---|---|---|---|
| First Income Decile | 114 | 1.13 | 1.13 |
| Second Income Decile | 288 | 2.86 | 4.00 |
| Third Income Decile | 605 | 6.02 | 10.01 |
| Fourth Income Decile | 621 | 6.17 | 16.19 |
| Fifth Income Decile | 1,012 | 10.06 | 26.25 |
| Sixth Income Decile | 1,023 | 10.17 | 36.42 |
| Seventh Income Decile | 1,101 | 10.95 | 47.37 |
| Eighth Income Decile | 1,434 | 14.26 | 61.63 |
| Nineth Income Decile | 1,925 | 19.14 | 80.77 |
| Tenth Income Decile | 1,934 | 19.23 | 100.00 |

Then, we use the formula:

$$\text{Gini} = \frac{n-1}{n} - \frac{2}{n} \times (s_1 + s_2 + \cdots + s_9),$$

where $s_i$ are the comulative shares and $n = 10$. The coefficient is therefore:

$$\text{Gini} = 0.9 - 0.2 \times (0.0113 + 0.04 + 0.1001 + 0.1619 + 0.2625 + 0.3642 + 0.4737 + 0.6163 + 0.8077) = 0.33$$

## 2.6 Measure of Poverty

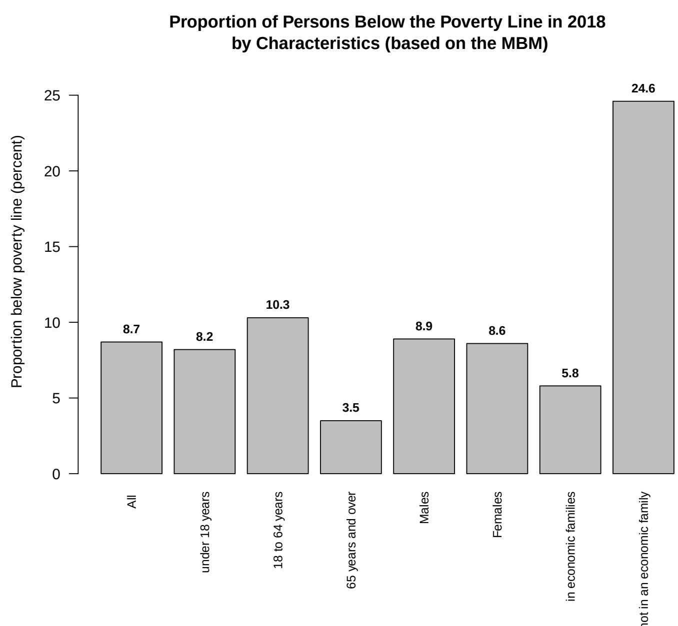
Income inequality is particularly problematic when it implies that a large proportion of the population is in a state of poverty. How do we define poverty and how do we measure it is the topic of this section. An household is considered poor if its income falls below the poverty line. Two of the measures used by Statistics Canada are the market basket measure and the low income measure. In this section, we use the Statistics Canada table 11-10-0135-01 (Statistics Canada, 2020f)

### The Market Basket Measure

The Statistics Canada definition of the market basket measure (MBM) is (Footnote 7 of the table):

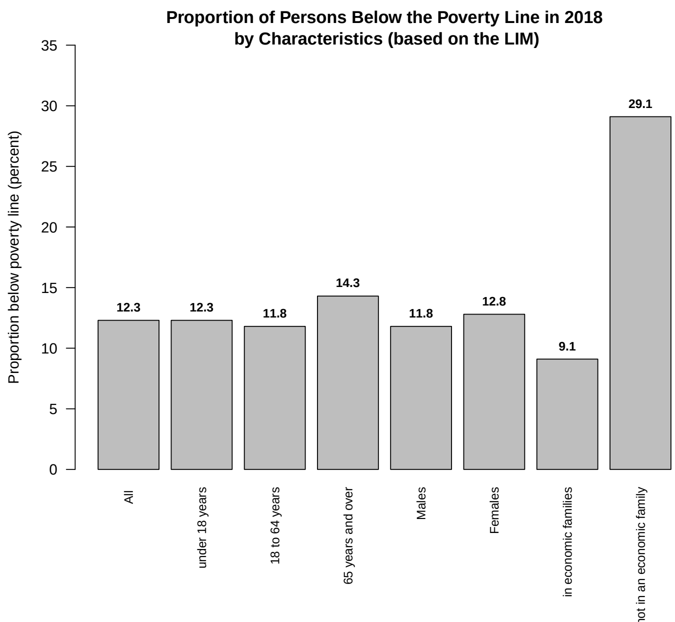
"The Market Basket Measure (MBM) is based on the cost of a specific basket of goods and services representing a modest, basic standard of living. It includes the costs of food, clothing, shelter, transportation and other items for a reference family. These costs are compared to the disposable income of families to determine whether or not they fall below the poverty line"

Therefore, an household is considered to be below the poverty line if they cannot afford the "modest and basic standard of living" determined by the basket. The table includes the proportion of persons that are below that poverty line by characteristics (sex, age, etc.) starting in 2006. The following bar chart shows the proportions for some groups in 2018. We can see that 24.6% of the persons who do not live in an economic families (a group of two and more persons living in the same dwelling) are below that poverty line. It is clearly the group of persons who is the most affected by poverty. Also, the group with the smallest proportion of persons below the poverty line is the 65 years and over, followed by the persons living in an economic families.



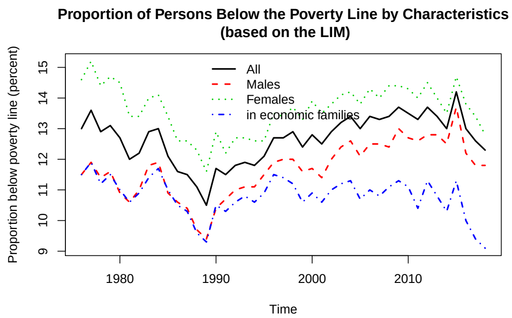

### The Low Income Measure

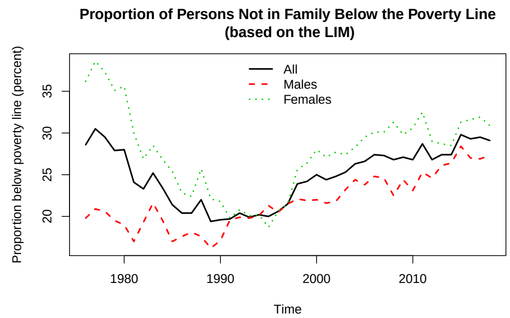
Another definition of poverty line is the low income measure (LIM). It is equal to half the adjusted median household income (adjusted for household size). In a previous section, we saw that the median household income in 2018 was $61,400. According to LIM, households with income less than $30,700 are below the poverty line. However, this household income is not adjusted for household sizes. The LIM from the Statistics Canada table 11-10-0135-01 is based on the adjusted after-tax income (can be selected in the Low income lines tab f the table). The following chart shows the proportion of persons below the poverty line based on the LIM using the same groups of persons. First, the proportion of persons below the poverty line in each group is higher with that definition. We do, however, observe similar differences across groups with one exception: the group 65 years and over is the second most affected group by poverty when the poverty line is based on the LIM and it is the least affected when it is based on the MBM. What is the best definition of poverty? That's a difficult question we won't answer in this course. What is important is that they both provide similar information about how many persons are at the bottom of the income distribution.




### The Poverty Over Time in Canada

The following graphs show the evolution of poverty in Canada from 1976 to 2018 based on the LIM. The left graph show the evolution for all person (black solid line), males (dashed red line), females (dotted green line) and persons in economic families (dot-dashed blue line) and the right graph shows the evolution for all persons not in family (black solid line), males not in family (dashed red line) and females not in family (dotted green line). We separated persons in family from persons not in family for clarity.




The overall conclusion is that poverty has decreased from 1976 to 1990 and increased after 1990. The only exception is for the persons in family for whom poverty has been quite stable over the whole period. This means that the overall changes in poverty is mostly driven by the persons not in family. If we compare the situations of males and females, we can see that the proportion of females below the poverty line has been systematically higher than the proportion of males throughout the period. However, the gap is much smaller in 2018 than in 1976, especially for persons not in family: 35% of females in 1976 compared with 20% of males and just a few percentage points difference in 2018.

**Important** Is poverty related to inequality? The simple answer is not necessarily, because we may have a situation in which everyone is poor. In this situation, there is no inequality. Similarly, we may have a situation in which the poorest individual can afford the MBM and so no individual is below the poverty line. But, this does not prevent a large portion of the population to be much richer then others. However, the relationship between poverty and inequality is more complicated than that. For this course, we will treat poverty and inequality as two different issues that may or may not be related.

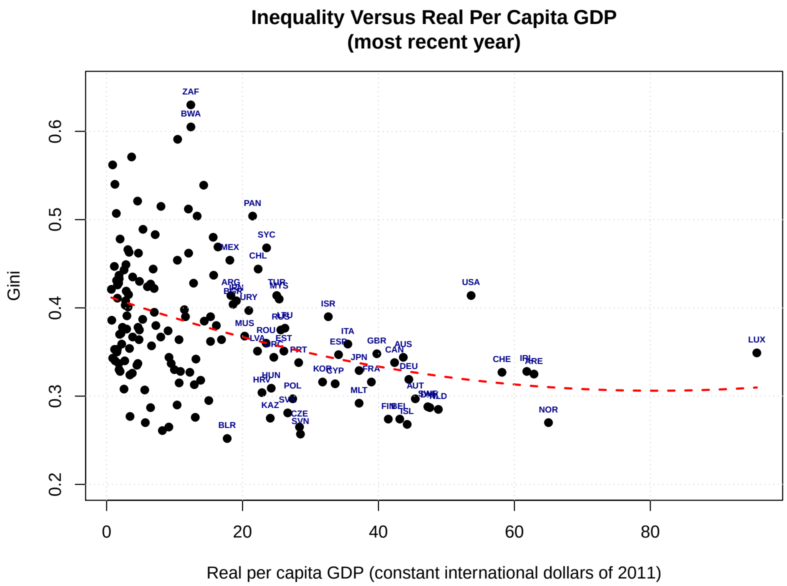
## 2.7 Inequality Versus Growth

In the module readings, you have read about theories that try to explain the possible relationships between inequality and growth. In this course, we do not have the theoretical background to understand those theories. We only want to focus on the empirical evidence. For example, you will read about the Kuznets hypothesis that predicts that there is a reversed U-shape relationship between inequality and per capita real GDP: it starts by being positively related and then becomes negatively related. Testing such relationships requires advanced statistical techniques. All we can do in this course is to verify if we observe such relationship. The data used in this section is from the World Bank (World Bank, 2020). You can explore the website and download the data by yourself or use the file worldData.csv available on the course website.

The data file worldData.csv contains 8 columns: the country names, the country codes, the most recent year when most variables were available, the real per capita GDP in thousands of international dollars of 2011, the Gini coefficients based on after-tax income, the income share of the bottom 10% of the income distribution, the income share of the top 10% of the distribution and the proportion of the population that are below the poverty line. The unit of the real GDP (international dollars of 2011) is a constant price measure that makes values from different countries comparable. We will learn more about this in a future module.

### Gini Versus Real Per Capita GDP Across Countries

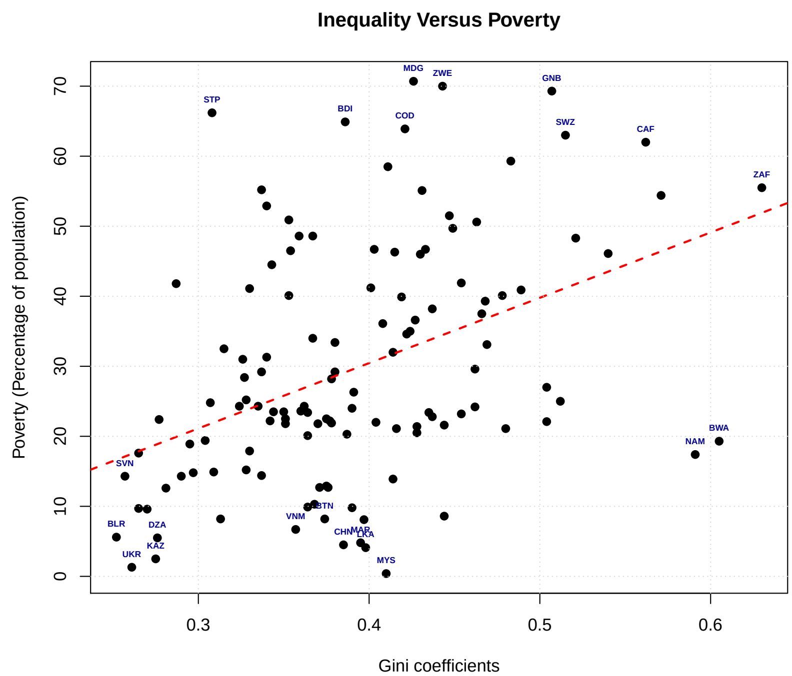
First, we want to see if we observe a relationship between real per capita GDP and inequality. The following is the scatter plot of the Gini as a function of real per capita GDP for the 154 countries included in the data file for which the two values are available1. For some selected points, the country codes are written on top. If you are not sure which country corresponds to the codes, type “iso 3166-1 alpha-3 codes” in any web search engine and you will find a list.




The read dashed line is just to show the average comovement between inequality and real per capita GDP. We observe a negative relationship between the Gini coefficient and real per capita GDP which suggest that richer countries are more equal on average. However, this is only true on average because there is a lot of variation. First, we see that the United States (USA) is more unequal than the average (about 68% of the countries have a smaller Gini coefficient than the US) even if it is one of the richest countries. Also, if we look at the poorest countries (20 thousand and below), we see that the Gini coefficient is all over the place. The least equal is Belarus (BLR) with a Gini coefficient of 0.252, and South Africa (ZAF) and Botswana (BWA) are the most unequal with Gini coefficients of 0.63 and 0.605 respectively. Yet, their per capita DGPs are not that different. Between them, we see about 108 countries with very different Gini coefficients. This result shows that poverty and inequality are not always related.

### Poverty Versus Inequality

The following is the scatter plot of the proportion of the population below the poverty line as a function of the Gini coefficient. In the data file, this measure of poverty is only available for 125 countries that are relatively poor (real per capita GDP less than 29 thousand). The red dashed line shows a positive comovement between the two variables on average. Therefore, there are fewer poor in more equal countries on average. However, that relationship is not true for many countries. For example, the proportion of poor in Ukraine is 1.3% and it is equal to 66.2% in São Tomé and Príncipe (STP), but the two countries have a relatively low Gini coefficient. On the right side of the graph, we have 17% of poor in Namibia (NAM) and 50% in South Africa (ZAF), and both countries have similar level of inequality.




### Poverty Versus Per Capita GDP

It is sometimes suggested that growth is the solution to the problem of poverty. If that's that case, government policies that promote growth should also help reducing poverty. In this section, we can to see if we observe a relationship between real per capita GDP and poverty. In the module readings, there is a scatter plot showing the relationship between the average income for the bottom quintile and real per capita GDP. The graph shows that there is a strong positive comovement between the two variables, which means that the poor get richer as the economy is growing. We can start this section by reproducing this graph. We need to compute the average income from the first decile using the information contained in our data file. The average income in decile i is given by the following formula:

$$\text{Average income in decile i} = 10 \times (\text{Real per capita GDP}) \times (\text{income share of decile i}) \,.$$

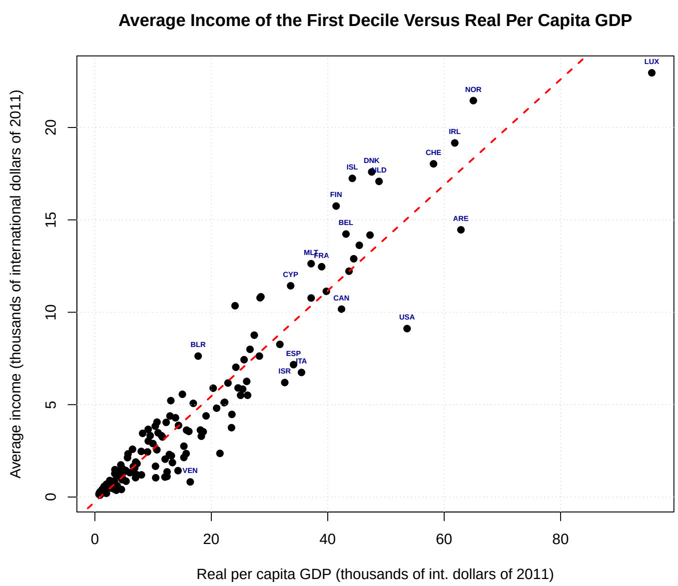
**Background Information** Let $I$ be the total income in the economy, Pop be the population and $I_i$ be the total income in the i$^{\text{th}}$ decile. Then, the average income in decile i is:

$$\text{Average income in decile i} = \frac{I_i}{Pop/10} = 10 \times \frac{I_i}{Pop}\,,$$

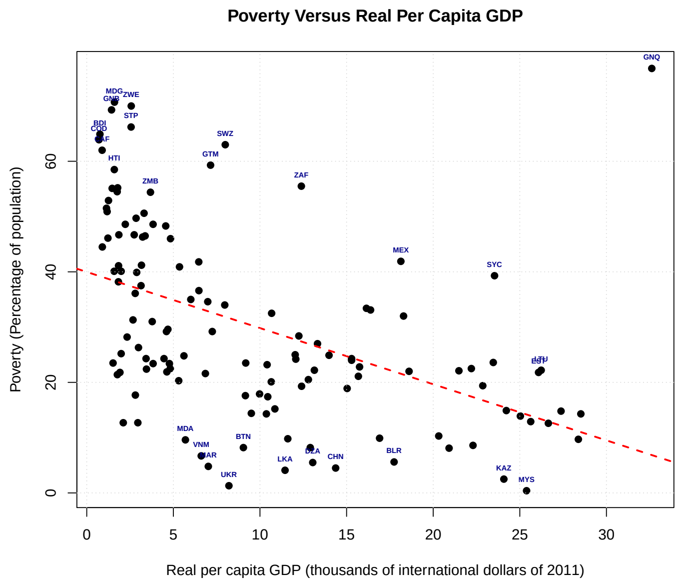
because the population of each decile is 1/10 of the total population. Using the fact that per capita GDP is $I/Pop$ and the income share of decile i is $I_i/I$, then

$$\begin{aligned} 10\times(\text{Real per capita GDP})\times(\text{income share of decile i}) &= 10\times\frac{I}{Pop}\times\frac{I_i}{I} \\ &= 10\times\frac{I_i}{Pop} \\ &= \text{Average income in decile i} \end{aligned}$$

Because it is based on GDP, it is equal to the average total income, not the average after-tax income. Our data file contains the income share and the real per capita GDP, so we can compute the average income from the first decile using the formula. The following is the scatter plot of the two variables. It is clear from the graph that higher average income implies higher average income for the bottom 10% of the income distribution. However, the relationship is not perfect: just compare the United States (USA) with Denmark (DNK) and Venezuela (VEN) with Belarus (BLR) (same average income but very different average income for the bottom decile).




The following graph compares directly poverty with real per capita GDP using the proportion of the population below the poverty line. On average (see the red dashed line), the proportion of the population below the poverty line is lower for countries with higher real per capita GDP. However, the relationship is not as clear as in the previous graph. Just compare Eswatini (SWZ) with Ukraine. They both have the same average income but proportion of poor in Eswatini is 63% and it is 1.3% in Ukraine. There is one important factor worth noticing: the variation gets smaller as average income increases. It implies that the number of countries with the same average income and very different proportion of poor gets smaller. This result does suggest that as real per capita GDP increases, the proportion of poor becomes low and stable.



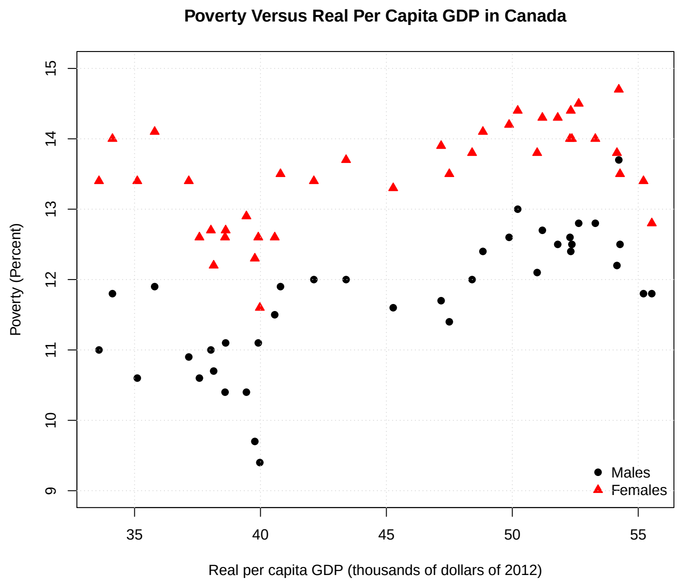

## 2.8 Exercise: Poverty versus Per Capita GDP in Canada

For this exercise, you need the proportion of persons below the poverty line (based on the LIM) by characteristics that we used in a previous section. The series are available in the Statistics Canada table 11-10-0135-01. You also need the annual real GDP (table 36-10-0222-01) and the annual population (table 17-10-0005-01) series.

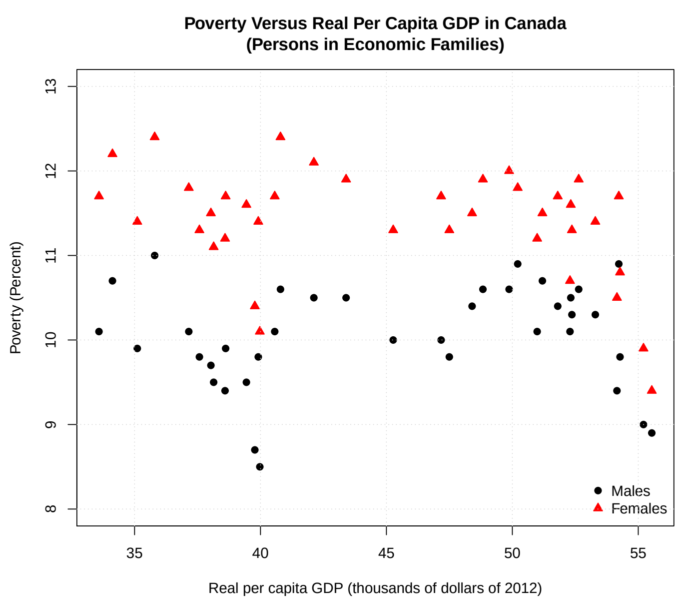
In the exercise, try to analyze the relationship between poverty and real per capital GDP by characteristics. We want to see if the increase in average income resulted in a reduction in poverty. To analyze the relationship, use scatter plots and compare: Males and Females, Males and Females in economic families, and Males and Females not in economic families.

### Solution

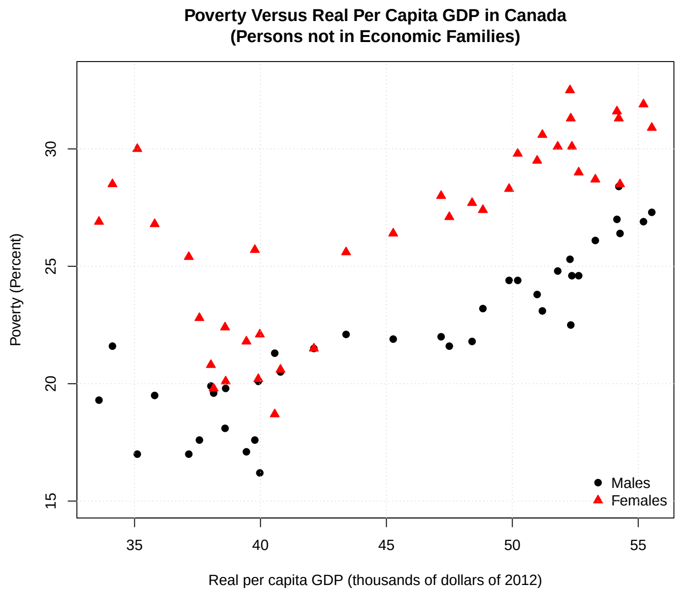
The following is the scatter plot of the proportion of persons below the poverty line (LIM) with respect to the real per capita GDP. The males are represented by the black circles and the females by the red triangles. For both series, it seems that higher average income is associated with a larger proportion of poor.




The following is the scatter plot of the proportion of persons below the poverty line (LIM) with respect to the real per capita GDP for persons in economic families. The males are represented by the black circles and the females by the red triangles. For males, there is no clear relationship between poverty and average income and for females, higher average income is associated with a lower proportion of poor. We also observe lots of variation: same average income associated with different level of poverty.




The following is the scatter plot of the proportion of persons below the poverty line (LIM) with respect to the real per capita GDP for persons not in economic families. The males are represented by the black circles and the females by the red triangles. For males, there is clear positive comovement between the two variables. For females, we see a negative relationship before real per capita GDP reached 40 thousand dollars of 2012 and a positive relationship after. Overall, however, it is a positive relationship that we observe. Therefore, growth does not seem to have solved the problem of poverty for that group.




## 3 The Sources and Consequences of Inequality

In this section, we summarize and briefly explain the important points that you need to remember from the module readings. It is not a substitute to reading, but a complement to help you identify the important results.

### Sources of Inequality

The following enumerate a few sources of inequality.

- Inequality in skills: If some workers have a particular skill that is highly in demand, these workers will be well paid compared tp the average workers. For example, skilled hockey players will be drafted by teams form the national hockey league (NHL) and will be paid much more than average workers. As another example, individuals with coding skills (software development, data science, etc.) are in high demand and therefore well paid compared with unskilled workers. This is a natural source of inequality that can hardly be controlled.
- Inequality in how skills are valued: Depending on how the society values products and services, some skills may result in higher income than others. It also depends on the demand and supply of those skills: if a particular skill is in high demand and low supply, it will be more valued.
- Inequality in education: We assume here that higher education leads to higher income. The following compares the median wage by level of education in 2016 (Statistics Canada, 2017).

**Median annual earnings of women and men aged 25 to 64 who worked full time and full year as paid employees**

| | High school diploma | Apprenticeship certificate | College diploma | Bachelor's degree |
|---|---|---|---|---|
| Women | $43,254 | $38,230 | $48,599 | $68,342 |
| Men | $55,774 | $72,955 | $67,965 | $82,082 |

From the table, more educated workers earn more: a gap of about $25,000 between high school diploma and bachelor's degree. Therefore, inequality of education is likely to create income inequality. Notice that education inequality across countries is also a source of income inequality in the world. Therefore, government policies that focus on reducing education inequality is likely to also reduce income inequality.

- Technological advances: This source is related to skills and education. For example, advances in information technology increases the value of computer scientists without affecting the value the unskilled workers or other type of unrelated skilled workers. For example, we saw in previous section that the Gini coefficient in Canada increased substantially in the '90s, a period characterized by a boom in computer technology.
- Value of education: Everything that affects the value of being educated will increase the impact of education inequality on income inequality. For example, technological advances affect the value of some types of education.

This is a short list of what may cause income inequality. We will cover other sources in more details in the Growth and Development module.

### The Effect of Inequality on Growth

Explaining the effect of inequality on growth requires theoretical background that we don't have, and testing empirically whether this relationship exists is something difficult even for well trained economists. The following quote from the readings summarizes well the issue:

"Does inequality raise or lower growth? Unfortunately, available statistical data are unable to answer this question. Although some economists claim to find evidence that inequality is on average bad for growth, others claim the data point in the opposite direction."

Just to mention the possible channels through which inequality may have an effect of growth, the inequality may affect:

- investment, which affects growth through its impact on the accumulation of capital goods (goods used to produce),
- education, which affects the productivity of workers,
- taxation (for redistribution purposes), which effects incentives to work or produce,
- the political stability, which affects the incentive to invest, and
- the level of criminality, which implies wasting resources to prevent it instead of producing goods.

Most economists think that those channels exist, but we don't know which one has the largest impact. For this course, you need to understand the intuition behind the different channels, and be aware of the difficulty of evaluating their relative impact.

> **Important** Only read the sub-section Empirical Evidence from the Effect of Income Inequality on Economic Growth section of the module readings. In particular, the historical evidence from Latin America, the United States and Canada at the end of the sub-section is a clear and intuitive description of how inequality may lead to lower growth.

You may read the other sections for own interest, but it is not required for the course. They require too much theoretical background.

**Economic Mobility**

Economic mobility is defined as the ability for individuals to move from one part of the income distribution to another (e.g. moving from the first quintile to the second of third quintile). This is much harder to measure than inequality. In the last section of the module readings, we see a transition matrix which is a way to measure mobility. For example, we see that persons whose parents were in the bottom quintile of the income distribution have 42% chance of being in the bottom decile, 23% chance of being in the second and 6% chance of being in the top decile. Also, persons whose parents are in the third quintile have almost equal chance to be in any quintile. This is a sign that there is some mobility in the economy. One main advantage of living in a highly mobile economy is that every one has equal chance of being successful. In absence of mobility, poor families would remain poor for generations.

Two important factors that affect mobility are:

- The accessibility of education: Children whose parents are poor have more chance to be in higher deciles if education is easily accessible.
- The accessibility of health care: Health has an impact on school performance. Therefore, having an accessible health system helps poor children to acquire a higher level of education and move to a higher income decile than their parents.

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